THE QUESTION

Changing the manager doesn't work the way you think.

A data analysis of 14 seasons of Bundesliga results (2012–2025), testing whether mid-season manager changes produce a real improvement — or whether the 'bounce' is mostly regression to the mean.

108
matched change events

1,963
control candidates

14
Bundesliga seasons

THE QUESTION THIS ANSWERS

Does changing the manager during a slump produce a real improvement — or would the team have recovered anyway?

WHAT 'MANAGER CHANGE' MEANS

The data detects when the manager name changes between matches. That includes dismissals, resignations, and caretaker handovers — these can't be separated without a reason-for-change field.

WHY THIS MATTERS

When a team struggles, the manager goes.

It's the default fix — in football, and in any organisation.

But teams in a slump often recover on their own, whether or not the leader changes.

THE ATTRIBUTION PROBLEM

01 **Teams in a slump recover on their own.**
Most of the post-change improvement happens regardless of the decision.

02 **The naive +0.64 PPG is misleading.**
It compares 'after change' to 'before change' — ignoring the control group.

03 **A matched comparison is the only honest test.**
108 change events vs. matched clubs in identical slumps that kept their manager.

So how much of the 'turnaround' is really the change?

1■■ THE NUMBER EVERYONE BELIEVES

+0.64

Points-per-game improvement in the 10 games after a manager is changed

WHY IT'S MISLEADING

This number compares a team's PPG after the change to its PPG before. But teams in a slump also recover naturally — with or without a new manager. Without a control group, the number tells you nothing about causation.

THE SLUMP THRESHOLD

PPG ≤ 1.0

over prior 10 matches — locked before computing outcomes

This is the number that gets cited. It's also misleading.

2■■ WHAT THAT NUMBER HIDES

Both groups were in an identical slump. The only difference: whether the manager was changed.

108 treated events matched to 1,963 control candidates on PPG and league position.

+0.64
CHANGED

+0.53
KEPT THE MANAGER



Struggling teams recover either way. Most of the 'bounce' isn't the new manager.

3■■ THE REAL NUMBER

+0.12

The actual gap linked to the manager change

— and it's not statistically significant at 10 matches

10-MATCH WINDOW

+0.12

p = 0.06 — not significant

5-MATCH ROBUSTNESS

+0.23

p = 0.005 — significant ✓

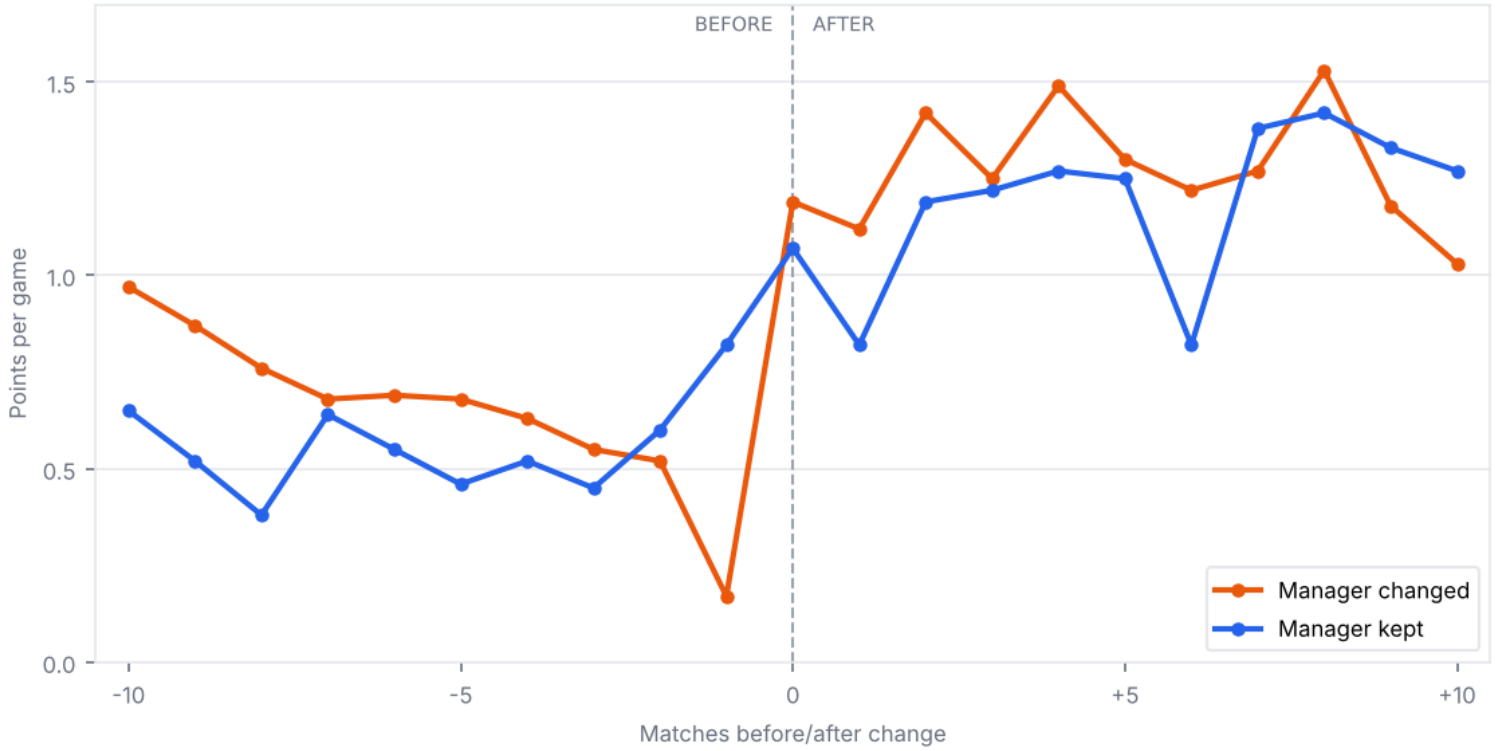
Suggests a short-lived front-loaded effect

108 matched cases. p = 0.06.

4■■ THE SHAPE OF THE BOUNCE

Most of the rebound is clubs recovering from their worst point.

Clubs change manager at their lowest point. The changed clubs sit slightly higher in the first few matches; the gap narrows over the full window.



■ **KEY FINDING**

A smaller extra lift remains for clubs that changed manager: +0.12 PPG over 10 matches (p = 0.06) and +0.23 over 5 (p = 0.005). A short, early new-manager effect can't be ruled out.

THE ANALYST

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AI & Analytics in HR · Solving People Problems across the Employee Lifecycle

Built this analysis to test a widely-cited stat with rigorous matched-comparison design.
Not an academic. Open to feedback.

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In-season manager changes detected

Bundesliga, 2012–2025

Robustness check:

5-match window → gap +0.23

(p = 0.005) ✓ significant

FULL ANALYSIS

01 **SQL build scripts, Python verification notebooks,**

02 **Power BI event-study source file,**

03 **Matched pairs CSV (108 treated events + controls).**

github.com/tarifhassan27/manager-bounce